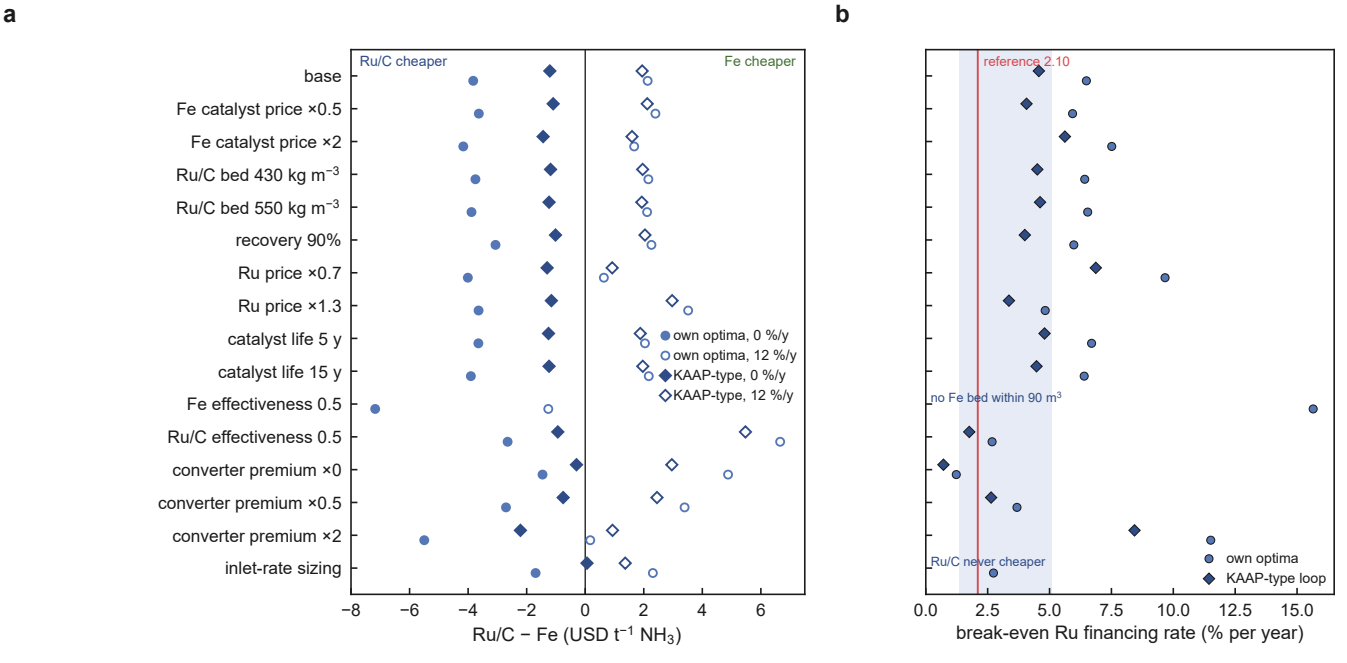
a
Ru/C cheaper
Fe cheaper
base
Fe catalyst price ×0.5
Fe catalyst price ×2
Ru/C bed 430 kg m−3
Ru/C bed 550 kg m−3
recovery 90%
Ru price ×0.7
Ru price ×1.3
catalyst life 5 y
catalyst life 15 y
Fe effectiveness 0.5
Ru/C effectiveness 0.5
converter premium ×0
converter premium ×0.5
converter premium ×2
inlet-rate sizing
own optima, 0 %/y
own optima, 12 %/y
KAAP-type, 0 %/y
KAAP-type, 12 %/y
−8
−6
−4
−2
0
2
4
6
Ru/C − Fe (USD t−1 NH3)
b
reference 2.10
no Fe bed within 90 m3
Ru/C never cheaper
own optima
KAAP-type loop
0.0
2.5
5.0
7.5
10.0
12.5
15.0
break-even Ru financing rate (% per year)